Project Charter: SnapFridge

Field	Details
Course	ISE304E – Software Engineering (2026–2027 Fall)
Group	SnapFridge – Group 2
Customer / Instructor	Res. Asst. Dr. Abdullah Cihan AK
Repository	https://github.com/iemiral-spec/SnapFridge
Document version / date	2.0 – 2026-11-03

1. Project Description

SnapFridge is a web application that helps users decide what to cook with the ingredients they already have. The user uploads a photo of their fridge or kitchen counter, an AI vision service marks the ingredients it recognizes on a list of common ingredients, the user corrects the selection or picks ingredients manually, and the system suggests recipes from a recipe database. The domain is food and lifestyle; the main users are students, employees and home cooks.

2. Objectives and Outcomes

Objectives

- Turn a fridge photo into an editable ingredient list in under 30 seconds.
- Suggest recipes ranked by how many of their ingredients the user already has, and show the missing ones.
- Deliver a cloud-hosted, tested web application that meets the course's technical requirements.

Outcomes (in scope)

- Registration, login and role-based authorization (User / Admin)
- Photo upload and AI-based ingredient recognition: one AI call per photo, results split into "sure" and "not sure"
- Ingredient selection screen with about 50 common ingredients and a search box; works with or without a photo
- Recipe dataset (TheMealDB; Kaggle as fallback) imported into PostgreSQL
- Ingredient-based recipe suggestions with recipe details and missing ingredients
- Favorite recipes
- Admin recipe management (add, edit, delete)

Optional (only if core work is finished): analysis history, recipe categories, keyword search, AI-generated recipes.

Out of scope: native mobile app, grocery ordering or payment, social features, calorie/diet tracking, training our own AI model.

Technology: Python FastAPI (REST API), PostgreSQL, JWT authentication, React + Tailwind CSS, Render (cloud hosting), GitHub Actions (CI), a vision-capable LLM API. AI coding assistants (Claude Code / GitHub Copilot) may be used; their work, effort, human review effort and the errors they introduce will be reported.

3. Project Manager and Staff

Member	Student ID	Role
Biray Dağlı	150210215	Project Manager + Backend Developer

Member	Student ID	Role
Ahmet İlker Yıldızlı	150210211	Configuration Manager + Frontend Developer
Umut Uğurlu	150230213	Test Manager + Backend Developer
Talha Arpacık	150220213	Documentation Lead + AI Integration Developer
İdil Emiral	150230223	Communication Lead + Database Developer

4. Summary Milestone Schedule

Dates follow the course's 13-week project outline (Wednesday classes, no class on 18 November). Report deadlines are 23:59 on the date shown; the exact time of each submission should be checked on Ninova.

Week	Date (2026)	Deliverable	Customer meeting / peer scoring
3	13 Oct	Project Charter	–
4	21 Oct	–	Meeting 1 on Project Plan
5	28 Oct	Project Plan (submitted 3 Nov)	–
6	4 Nov	–	Meeting 2 on Requirements
7	10 Nov	Requirements Specification Report	Peer Scoring 1 (11 Nov)
8	24 Nov	Requirements Analysis Report	Midterm (25 Nov)
9	2 Dec	–	Peer Scoring 2
10	8 Dec	Software Design Report	Meeting 3 on Design (9 Dec)
12	22 Dec	Software Test Report, Code Freeze	Meeting 4 on Tests (23 Dec)
13	30 Dec, 12:30	Codebase (project submission), Final Demo and Presentation	Peer Scoring 3

5. Preliminary Cost Estimation

- **Effort:** about 170 person-hours of development work (5 members × ~10 hours/week over 8 weeks, 2 November – 28 December, for project activities), plus time for reports and meetings.
- **Money:** no budget. We use free tiers and student offers: GitHub (Student Pack), Render free tier, PostgreSQL on Render, TheMealDB. The vision API is pay-per-use; we will pick a provider with a free tier or student credits and limit calls during development.

6. Preliminary Risks

Risk	Impact	Response
AI recognizes ingredients wrongly	Wrong recipe suggestions	AI chooses only from the common list; uncertain items shown separately; user always confirms
Vision API cost, quota or downtime	Main flow stops	One call per photo; users can still pick ingredients from the list without the AI
Recipe dataset licence or coverage is not adequate	Few or no matches	Switch to a Kaggle dataset or a smaller hand-picked set
Exams and other courses slow the team	Missed deadlines	Small core scope, weekly check-in, optional features dropped first
Scope grows after customer meetings	Schedule slips	New requests go to the optional list unless the customer approves a trade-off
Free hosting limits (sleep, storage)	Demo problems	Rehearse deployment early; keep a recorded demo as backup